SECSR2043 OPERATING SYSTEMS
[20 Marks]

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Section	:	03	

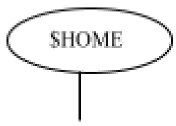
Instruction: Please answer all of the following questions. Whenever the □ symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname*.sh (Example: `ahmad.sh`).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd
../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.* .; cp ../fruits/*.* .;
cp ../*.jar .
```

a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **$HOME** directory.

[4 marks]



Print screen the script that you type;

```
syahmi@secr2043: ~

  GNU nano 7.2                          syahmi.sh
#!/bin/bash

echo "Hello world" > helloworld.jar

mkdir cars
mkdir dates
mkdir fruits drinks

cd cars
echo "Honda Accord" > accord.c
cp accord.c civic.c
echo proton > proton.c

cd ../dates
date > dateoftheday
cat dateoftheday > appointment

cd ../fruits
echo apple > apple.txt
cat apple.txt > orange.txt

cd ../drinks
cp ../cars/*.* .
cp ../fruits/*.* .
cp ../*.jar .

                              [ Read 25 lines ]
^G Help          ^O Write Out     ^W Where Is      ^K Cut           ^T Execute
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify
```

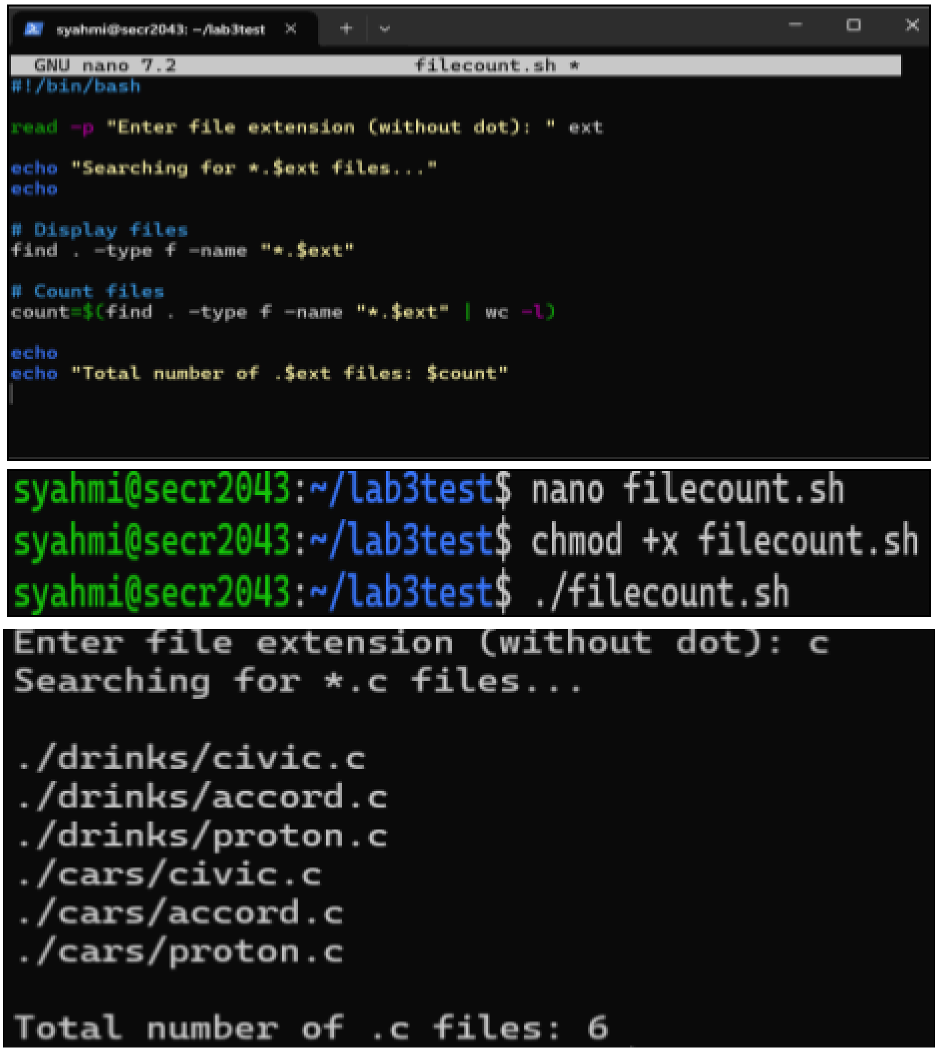
Then draw the tree

```
/home/syahmi/lab3test
|-- cars
|    |-- accord.c
|    |-- civic.c
|    `-- proton.c
|-- dates
|    |-- appointment
|    `-- dateoftheday
|-- drinks
|    |-- accord.c
|    |-- apple.txt
|    |-- civic.c
|    |-- helloworld.jar
|    |-- orange.txt
|    `-- proton.c
|-- fruits
|    |-- apple.txt
|    `-- orange.txt
|-- helloworld.jar
`-- syahmi.sh

5 directories, 15 files
```

b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter "c" as the input to the bash script. [4 marks]

Print screen the bash script you type and run



```
syahmi@secr2043: ~/lab3test
GNU nano 7.2                    filecount.sh *
#!/bin/bash

read -p "Enter file extension (without dot): " ext

echo "Searching for *.$ext files..."
echo

# Display files
find . -type f -name "*.$ext"

# Count files
count=$(find . -type f -name "*.$ext" | wc -l)

echo
echo "Total number of .$ext files: $count"
```

```
syahmi@secr2043:~/lab3test$ nano filecount.sh
syahmi@secr2043:~/lab3test$ chmod +x filecount.sh
syahmi@secr2043:~/lab3test$ ./filecount.sh
```

```
Enter file extension (without dot): c
Searching for *.c files...

./drinks/civic.c
./drinks/accord.c
./drinks/proton.c
./cars/civic.c
./cars/accord.c
./cars/proton.c

Total number of .c files: 6
```

2. The following Figure 1 illustrates a tree structure of some directories and files.

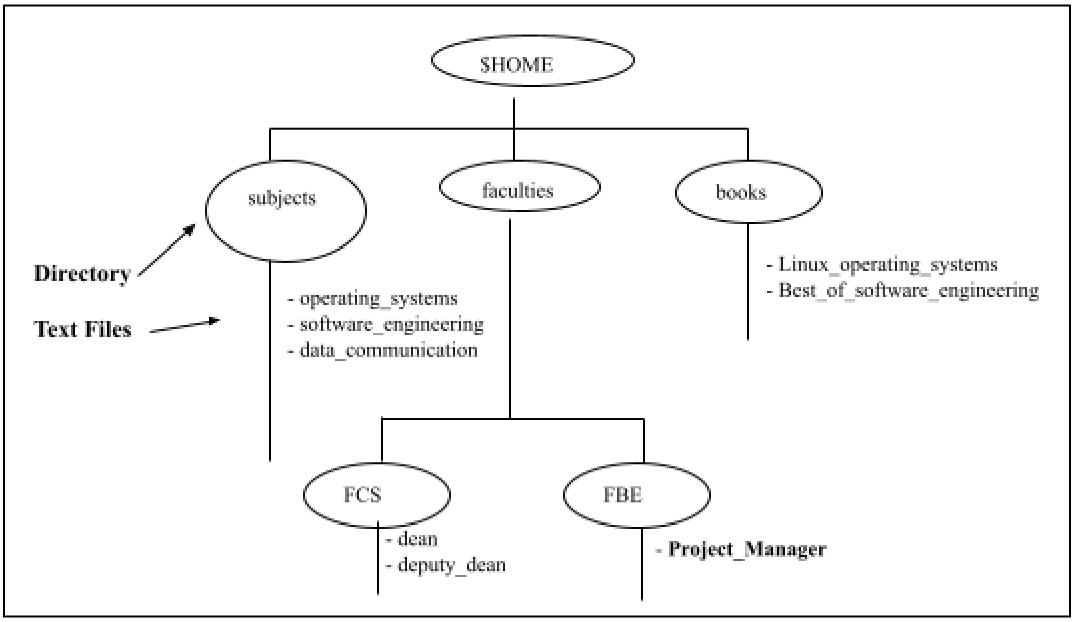
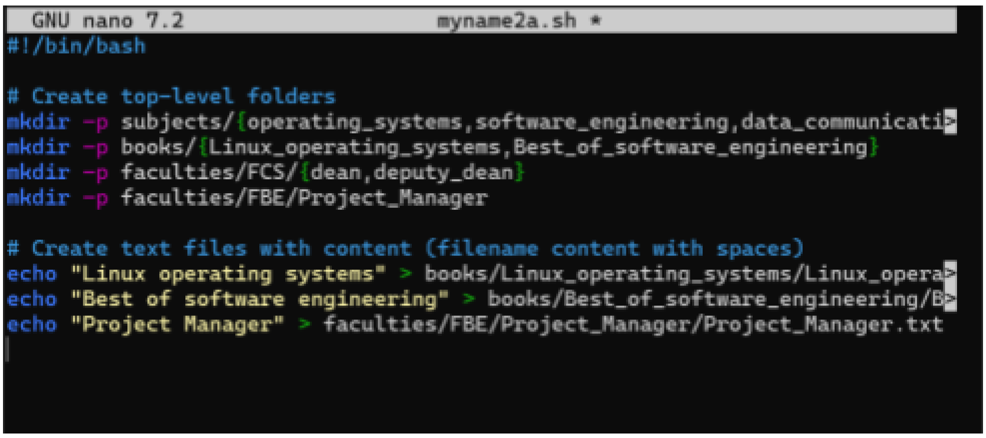


Figure 1

a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run



```
GNU nano 7.2                    myname2a.sh *
#!/bin/bash

# Create top-level folders
mkdir -p subjects/{operating_systems,software_engineering,data_communicati>
mkdir -p books/{Linux_operating_systems,Best_of_software_engineering}
mkdir -p faculties/FCS/{dean,deputy_dean}
mkdir -p faculties/FBE/Project_Manager

# Create text files with content (filename content with spaces)
echo "Linux operating systems" > books/Linux_operating_systems/Linux_opera>
echo "Best of software engineering" > books/Best_of_software_engineering/B>
echo "Project Manager" > faculties/FBE/Project_Manager/Project_Manager.txt
```

```
syahmi@secr2043:~/lab3test$ nano myname2a.sh
syahmi@secr2043:~/lab3test$ chmod +x myname2a.sh
syahmi@secr2043:~/lab3test$ ./myname2a.sh
syahmi@secr2043:~/lab3test$ tree ~/lab3test
/home/syahmi/lab3test
|-- books
|   |-- Best_of_software_engineering
|   |   `-- Best_of_software_engineering.txt
|   `-- Linux_operating_systems
|       `-- Linux_operating_systems.txt
|-- cars
|   |-- accord.c
|   |-- civic.c
|   `-- proton.c
|-- dates
|   |-- appointment
|   `-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   `-- proton.c
|-- faculties
|   |-- FBE
|   |   `-- Project_Manager
|   |       `-- Project_Manager.txt
|   `-- FCS
|       |-- dean
|       `-- deputy_dean
|-- filecount.sh
|-- fruits
|   |-- apple.txt
|   `-- orange.txt
|-- helloworld.jar
|-- myname2a.sh
|-- subjects
|   |-- data_communication
|   |-- operating_systems
|   `-- software_engineering
`-- syahmi.sh

18 directories, 20 files
```

b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called book. [2 marks]

Directory/File	Access Control
books	drwxr-xr-x
subjects	drwxr-xr-x
Best_of_software_engineering	drwxr-xr-x
FCS	drwxr-xr-x
project_manager	drwxr-xr-x

```
syahmi@secr2043:~/lab3test$ pwd
/home/syahmi/lab3test
syahmi@secr2043:~/lab3test$ ls -ld books subjects books/Best_of_software_engineer
ing faculties/FCS faculties/FBE/Project_Manager
drwxr-xr-x 4 syahmi syahmi 4096 Jun 16 11:38 books
drwxr-xr-x 2 syahmi syahmi 4096 Jun 16 11:38 books/Best_of_software_engineering
drwxr-xr-x 2 syahmi syahmi 4096 Jun 16 11:38 faculties/FBE/Project_Manager
drwxr-xr-x 4 syahmi syahmi 4096 Jun 16 11:38 faculties/FCS
drwxr-xr-x 5 syahmi syahmi 4096 Jun 16 11:38 subjects
```

c) Write another bash script (called `myname2c.sh`) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
`subjects`	✓	✓	✓	✓	x	x	✓	x	x
`Best_of_software_engineering`	✓	x	✓	x	✓	x	x	x	x
`FCS`	✓	✓	x	x	x	x	✓	✓	✓
`project_manager`	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run

```
GNU nano 7.2                         myname2c.sh *
#!/bin/bash

# subjects: rwx for owner, x for group, x for public => 711
chmod 711 subjects

# Best_of_software_engineering: rwx for owner only => 700
chmod 700 books/Best_of_software_engineering

# FCS: rwx for owner, x for group, x for public => 711
chmod 711 faculties/FCS

# Project_Manager: x for group and public only => 011
chmod 011 faculties/FBE/Project_Manager
```

```
syahmi@secr2043:~/lab3test$ nano myname2c.sh
syahmi@secr2043:~/lab3test$ chmod +x myname2c.sh
syahmi@secr2043:~/lab3test$ ./myname2c.sh
```

d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subjects	drwx------
Best_of_software_engineering	d--x--x--x
FCS	drwx--x--x
project_manager	drwx--x--x

```
syahmi@secr2043:~/lab3test$ ls -ld subjects books/Best_of_software_engineering fa
culties/FCS faculties/FBE/Project_Manager
drwx------ 2 syahmi syahmi 4096 Jun 16 11:38 books/Best_of_software_engineering
d-----x--x 2 syahmi syahmi 4096 Jun 16 11:38 faculties/FBE/Project_Manager
drwx--x--x 4 syahmi syahmi 4096 Jun 16 11:38 faculties/FCS
drwx--x--x 5 syahmi syahmi 4096 Jun 16 11:38 subjects
```

End of Lab 3

**** All the Best for Final Exam ****